EECS 841: Computer Vision

Lecture 10

Instructor: Sumaiya Shomaji,
Assistant Professor,
Dept. of EECS, University of Kansas
Email: shomaji@ku.edu
Office: 2020 Eaton Hall and 251 Nichols Hall

Limitations of Point Operations

- They don’t know where they are in the image
- They don’t know anything about their neighbors
- Most image features (edges, textures, etc.) involve the use of a spatial neighborhood of pixels
- If we want to manipulate or enhance these features, we need to go beyond point operations.
- Things that point operators cannot do:
  - **Blurring or Smoothing**
  - **Sharpening**

Solution?

How can we enhance the features more?

Ans: Filtering (Spatial Filtering)

- A **spatial filter** is an image operation where each pixel value $I(u,v)$ is changed by a function of the intensities of pixels in a neighborhood of (u,v).

Definition

- A **spatial filter** is an image operation where each pixel value *I(u,v)* is changed by a function of the intensities of pixels in a neighborhood of *(u,v)*.

Spatial Filtering Methods

- Two types of Spatial Filtering (**Linear and Non-Linear**)
- Two types of Linear Spatial Filtering Method (**Correlation and Convolution**)

Importance of the Two Linear Spatial Filters

Correlation:

- Similar to matrix multiplication between image and kernel
- Application 1: Similarity measure ☾ object detection (ex. face)
- Application 2: Pre-processing ☾ smoothing, sharpening etc.

Convolution:

- Flipflop + Correlation
- Application 1: extracts relevant features (as kernel's) from input image **based on changes in the surrounding pixels** (changes are enhanced since we do flip flop = adding more challenging to the image = flaunting the changes)
- Application 2: Application 1 useful for knowledge guided learning (ex. NN)

Focus of this Lecture

Correlation:

- Similar to matrix multiplication between image and kernel
- Application 1: Similarity measure ☾ object detection (ex. face)
- Application 2: Pre-processing ☾ smoothing, sharpening etc.

Convolution:

- Flipflop + Correlation
- Application 1: extracts relevant features (as kernel's) from input image **based on changes in the surrounding pixels** (changes are enhanced since we do flip flop = adding more challenging to the image = flaunting the changes)
- Application 2: Application 1 useful for knowledge guided earning (ex. NN)

Focus of this Lecture

Correlation:

- Similar to matrix multiplication between image and kernel
- Application 1: Similarity measure ☾ object detection (ex. face)
- Application 2: Pre-processing ☾ smoothing, sharpening etc.

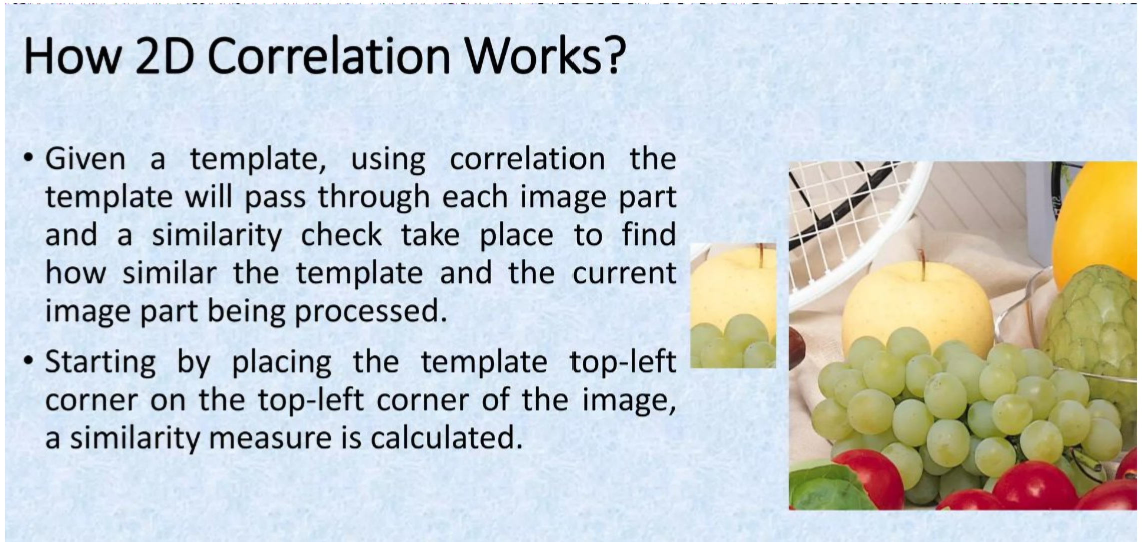
Convolution:

- Flipflop + Correlation
- Application 1: extracts relevant features (as kernel's) from input image **based on changes in the surrounding pixels** (changes are enhanced since we do flip flop = adding more challenging to the image = flaunting the changes)
- Application 2: Application 1 useful for knowledge guided earning (ex. NN)

Correlation: Introduction

Correlation for 2D Image

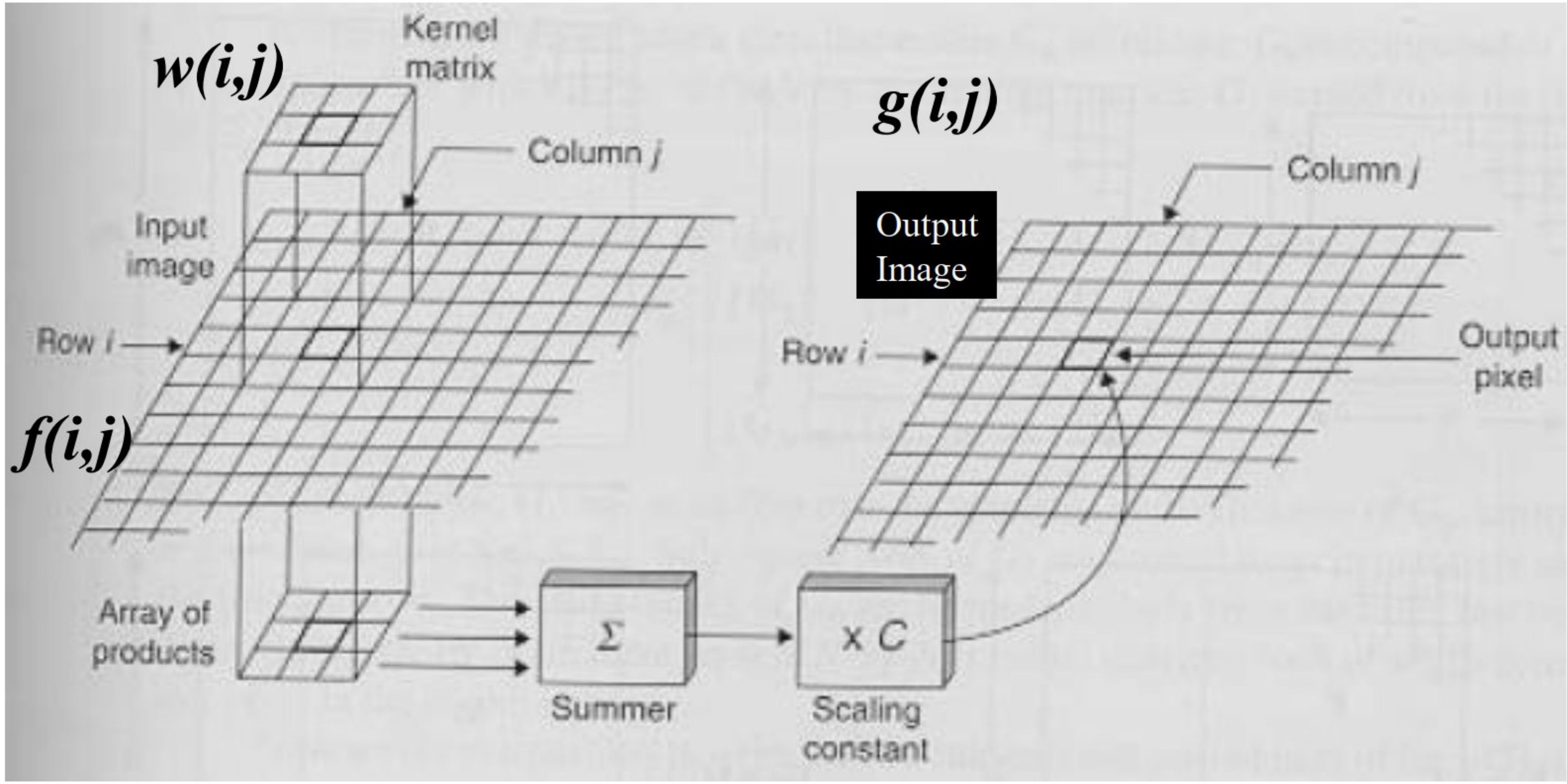
- Correlation is used to match a template to an image.
- Can you tell where the following template exactly located in the image?
- Using correlation we can find the image region that best matches the template.



How 2D Correlation Works?

- Given a template, using correlation the template will pass through each image part and a similarity check take place to find how similar the template and the current image part being processed.
- Starting by placing the template top-left corner on the top-left corner of the image, a similarity measure is calculated.

Correlation



$$g(i,j) = w(i,j) \bullet f(i,j)$$

Correlation

Correlation for 2D Image

1	3	4	3
2	7	4	1
6	2	5	2
13	6	8	9

Template – 3x3

2	1	-4
3	2	5
-1	8	1

Correlation

Correlation for 2D Image

1	3	4	3
2	7	4	1
6	2	5	2
13	6	8	9

Template – 3x3

2	1	-4
3	2	5
-1	8	1

Correlation for 2D Image

1	3	4	3
2	**7**	4	1
6	2	5	2
13	6	8	9

Template – 3x3

$$2 \ast 1 + 1 \ast 3 - 4 \ast 4 + 3 \ast 2 + 2 \ast 7 + 5 \ast 4 - 1 \ast 6 + 8 \ast 2 + 1 \ast 5 = \mathbf{44}$$

Correlation

Correlation for 2D Image

1	3	4	3
2	7	4	1
6	2	5	2
13	6	8	9

Template – 3x3

2	1	-4
3	2	5
-1	8	1

Correlation for 2D Image

1	3	4	3
2	**7**	4	1
6	2	5	2
13	6	8	9

Template – 3x3

$$
\begin{aligned}
&2 * 1 + 1 * 3 - 4 * 4 + 3 * 2 + 2 * 7 + 5 * 4 \\
&- 1 * 6 + 8 * 2 + 1 * 5 \\
&= \mathbf{44}
\end{aligned}
$$

output

	44		

Template – 3x3

Correlation

Correlation for 2D Image

1	3	4	3
2	7	4	1
6	2	5	2
13	6	8	9

Template – 3x3

2	1	-4
3	2	5
-1	8	1

1	3	4	3
2	7	4	1
6	2	5	2
13	6	8	9

$$2 * 3 + 1 * 4 - 4 * 3 + 3 * 7 + 2 * 4 + 5 * 1 - 1 * 2 + 8 * 5 + 1 * 2 = \mathbf{72}$$

Correlation

Correlation for 2D Image

1	3	4	3
2	7	4	1
6	2	5	2
13	6	8	9

Template – 3x3

2	1	-4
3	2	5
-1	8	1

output

	44	72	

Template – 3x3

Focus of this Lecture

Correlation:

- Similar to matrix multiplication between image and kernel
- Application 1: Similarity measure ☾ object detection (ex. fruit detect)
- Application 2: Pre-processing ☾ smoothing, sharpening etc.

Convolution:

- Flipflop + Correlation
- Application 1: extracts relevant features (as kernel's) from input image **based on changes in the surrounding pixels** (changes are enhanced since we do flip flop = adding more challenging to the image = flaunting the changes)
- Application 2: Application 1 useful for knowledge guided earning (ex. NN)

Correlation

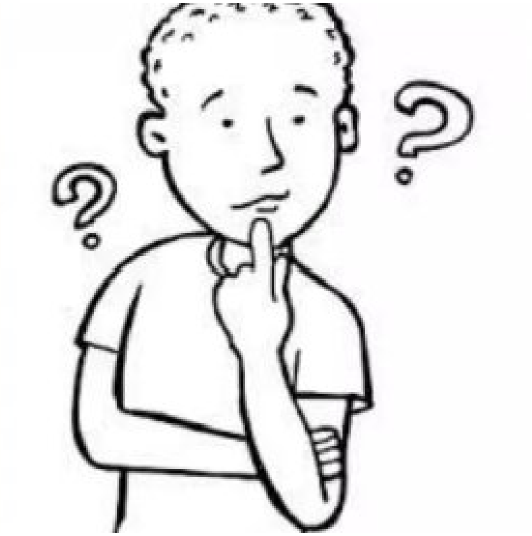
Based on the current step, what is the region that best matches the template?

output

	44	72	

It is the one corresponding to the highest score in the result matrix.

Special Note #1 in Correlation: Odd template



Special Note #1 in Correlation: Boundary

1	3	2	3	-4
2	7	3	2	5
6	2	1	8	1
13	6	8	9	

This will cause the program performing correlation to fall into index out of bounds exception.

Solution

Padding by Zeros

1	3	4	3	0
2	7	4	1	0
6	2	5	2	0
13	6	8	9	0

Continue

Correlation for 2D Image

1	3	4	3
2	7	4	1
6	2	5	2
13	6	8	9

Template – 3x3

2	1	-4
3	2	5
-1	8	1

Before zero pad

1	3	2	1	-4
2	7	3	2	5
6	2	-1	8	1
13	6	8	9	

After zero pad

1	3	2	1	-4
2	7	3	2	5
6	2	-1	8	1
13	6	8	9	0

$$2*4+1*3-4*0+3*4+2*1+5*0$$
$$-1*5+8*2+1*0$$
$$=\mathbf{36}$$

Continue

Correlation for 2D Image

1	3	4	3
2	7	4	1
6	2	5	2
13	6	8	9

Template – 3x3

2	1	-4
3	2	5
-1	8	1

Before zero pad

1	3	4/2	3/1	-4
2	7	4/3	1/2	5
6	2	5/-1	2/8	1
13	6	8	9	

After zero pad

1	3	4/2	3/1	0/-4
2	7	4/3	1/2	0/5
6	2	5/-1	2/8	0/1
13	6	8	9	0

$$2*4+1*3-4*0+3*4+2*1+5*0$$
$$-1*5+8*2+1*0$$
$$=\mathbf{36}$$

Continue

Correration for 2D Image

1	3	4	3
2	7	4	1
6	2	5	2
13	6	8	9

Template – 3x3

2	1	-4
3	2	5
-1	8	1

Before zero pad

1	3	4	3	-4
2	7	4	1	5
6	2	5	2	1
13	6	8	9	

After zero pad

1	3	4	3	0
2	7	4	1	0
6	2	5	2	0
13	6	8	9	0

output

	44	72	36

$$2*4+1*3-4*0+3*4+2*1+5*0$$
$$-1*5+8*2+1*0$$
$$=\mathbf{36}$$

One last demonstration

Correlation for 2D Image

1	3	4	3
2	7	4	1
6	2	5	2
13	6	8	9

Template – 3x3

2	1	-4
3	2	5
-1	8	1

1	3	4	3
2	7	4	1
6	2	5	2
13	6	8	9

How to overlay the kernel to find the corresponding value for 8?

One last demonstration

Correlation for 2D Image

1	3	4	3
2	7	4	1
6	2	5	2
13	6	8	9

Template – 3x3

2	1	-4
3	2	5
-1	8	1

1	3	4	3
2	7	4	1
6	2	5	2
13	6	8	9

How to overlay the kernel to find the corresponding value for 8?
And: by zero padding!

After zero pad

1	3	4	3
2	7	4	1
6	2	5	2
13	6	8	9
0	**0**	**0**	**0**

1	3	4	3
2	7	4	1
6	2	5	2
13	6	**8**	9
0	**0**	**0**	**0**

output

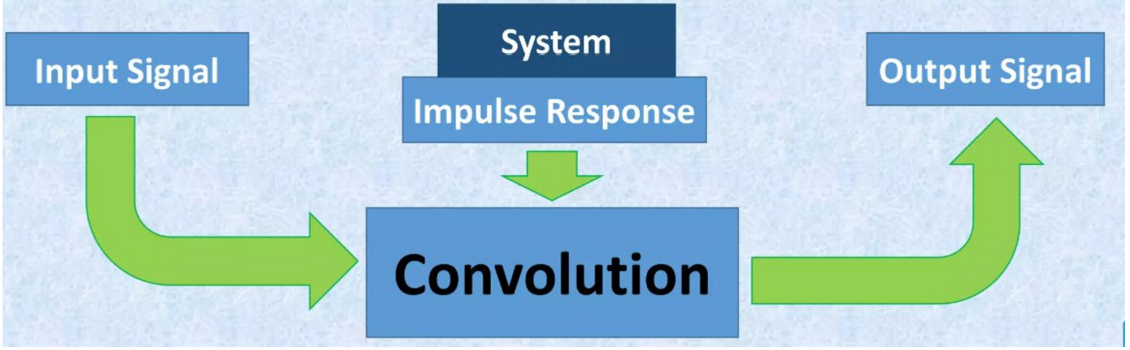
	44	**72**	**36**
		80	

$$2*2+1*5-4*2+3*6+2*8+5*9-1*0+8*0+1*0 = \mathbf{80}$$

Convolution

What is Convolution?

- The primary objective of mathematical convolution is combining two signals to generate a third signal. Convolution is not limited on digital image processing and it is a broad term that works on signals.



Convolution Applications

- Convolution is used to merge signals.
- It is used to apply operations like smoothing and filtering images where the primary task is selecting the appropriate filter template or mask.
- Convolution can also be used to find gradients of the image.

Operation

Applying convolution is similar to correlation except for flipping the template before multiplication.

Just rotate the template with 180 degrees before multiplying by the image.

1, 1	1, 0	1, -1
0, 1	0, 0	0, -1
-1, 1	-1, 0	-1, -1

-1, -1	-1, 0	-1, 1
0, -1	0, 0	0, 1
1, -1	1, 0	1, 1

Focus of this Lecture

Correlation:

- Similar to matrix multiplication between image and kernel
- Application 1: Similarity measure ☾ object detection (ex. face)
- Application 2: Pre-processing ☾ smoothing, sharpening etc.

Convolution:

- Flipflop + Correlation
- Application 1: extracts relevant features (as kernel's) from input image **based on changes in the surrounding pixels** (changes are enhanced since we do flip flop = adding more challenging to the image = flaunting the changes)
- Application 2: Application 1 useful for knowledge guided earning (ex. NN)

Smoothing Filter

Objective of Smoothing Filters:

- Blurring
- Noise reduction

Why are they needed:

- Blurring: preprocessing step/ image downsizing job ☾ object detection gets easier in a blurred/ less detailed image
- Noise reduction: preprocessing step ☾ ease the task of object detection or classification etc.

2 Types of Smoothing Filters:

- Linear
- Nonlinear

2 Types of Smoothing Spatial Filters

Linear Filters

- They mostly use averaging or lowpass filters
- They average the pixels in a neighborhood overlapped with the filter/kernel/mask
- Consequently, it reduces the effect of sharp features
- Thus, the smoothing is in effect as well as noise reduction occurs

- Mean/average/box filter
- Weighted average filter
- Gaussian filter

Non-Linear Filters

- They function by order or ranking of the pixels of a neighborhood of a certain area
- Center value finally gets replaced by the order/rank specified value

- Median filter
- Max filter
- Min filter

Smoothing Spatial Filters: Linear

- **Mean/average/box filter**

$$\frac{1}{9}\times\begin{bmatrix}1&1&1\\1&1&1\\1&1&1\end{bmatrix}$$

Filter = kernel = weighting mask

Every pixel gets equal weight

Example:

100	100	100	100	100
100	200	205	203	100
100	195	200	200	100
100	200	205	195	100
100	100	100	100	100

$$\frac{1}{9}\times\begin{bmatrix}1&1&1\\1&1&1\\1&1&1\end{bmatrix} \longrightarrow$$

100	100	100	100	100
100	144	167	145	100
100	167	200	168	100
100	144	166	144	100
100	100	100	100	100

Smoothing Spatial Filters: Linear

- **Weighted average filter**

  Considers unequal weighting to favor or prioritize nearby pixels

$$\frac{1}{16}\times\begin{array}{|c|c|c|}\hline 1 & 2 & 1\\\hline 2 & 4 & 2\\\hline 1 & 2 & 1\\\hline\end{array}$$

100	100	100	100	100
100	200	205	203	100
100	195	200	200	100
100	200	205	195	100
100	100	100	100	100

$$\frac{1}{16}\times\begin{array}{|c|c|c|}\hline 1 & 2 & 1\\\hline 2 & 4 & 2\\\hline 1 & 2 & 1\\\hline\end{array} \longrightarrow$$

100	100	100	100	100
100	156	176	158	100
100	174	201	175	100
100	156	175	156	100
100	100	100	100	100

Smoothing Spatial Filters: Linear

- **Gaussian filter**

Constructing Gaussian Kernels

- To construct a 1D Gaussian kernel with an arbitrary standard deviation σ, simply sample the continuous zero-mean Gaussian function:

$$gauss_{\sigma^2}(x) \equiv \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-x^2}{2\sigma^2}\right)$$

$\frac{1}{16} \times$

1	2	1
2	4	2
1	2	1

Smoothing Spatial Filters: Linear

Which one is Gaussian, and which one is average?

Smoothing Spatial Filters: Non-Linear

- **Median filter-** choose a neighborhood size ☾ find the median of the pixels from that neighborhood ☾ replace the center value with that

- **Min filter-** choose a neighborhood size ☾ find the minimum of the pixels from that neighborhood ☾ replace the center value with that

- **Max filter-** choose a neighborhood size ☾ find the maximum of the pixels from that neighborhood ☾ replace the center value with that

Practice problem on Smoothing Spatial Filters

Question: Consider the 5x5 image below. If you apply (a) – (e) filter considering 3 x 3 neighborhood, what will be the corresponding value of the pixel at (2,2) position considering this as the center pixel

2	6	7	8	9
4	7	9	5	1
5	4	6	8	3
4	2	0	1	2
0	8	8	9	0

(a) Average filter
(b) Weighted average
(c) Median filter
(d) Min filter
(e) Max filter